Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[utf8]inputenc [english]babel preamble [backend=biber,style=chem-acs]biblatex biblio.bib

reddy University of Pavia

Master's Degree Thesis

Dr. Bianchi's Thesis Template Explained

– directly on his template –

Author:
reddy Alessio Bianchi

Supervisor:
Your Supervisor's name

Co-Supervisor:
reddy Your Co-Supervisor's name

A thesis submitted in fullfillment of the requirements for the

reddy Master's Degree in Chemistry

in the

reddy Department of Chemistry
Organic Chemistry Section

July, 2022

1.2

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Bagian I

Using Parts is Not Mandatory, Actually

Bab 3

Probably, The Introduction

A cool citation is always an extra-point.– Someone Important 1.2

3.1 A Little Overview

This template has been directly made by modifying the common LaTeX's reddy `book` documentclass. You can find the related code, with comments, in the reddy `preamble.sty` file. The default font-family is reddy `Liunx Libertine/Biolinum` in which the second one – Sans variant – is set as default, while math environments like this

$$x(t) = A\sin(\omega t + \phi)$$

use the Serif variant: nobody forbids you from using different fonts (like the ones already included in the preamble or completely different).

3.1.1 The Color Accent

The document possesses a color accent called 'reddy `reddy`' which is used for chapter titles, footnotes numbers,[1] bullets in lists, and others. You can recall it to color-up your stuff with reddy `\color{reddy}` and modifying its RGB color definition in the preamble file also.

3.1.2 The `preamble.sty` File

The preamble file contains adjustments in margins, tables, headers and footers, and a lot of other stuff: the best piece of advice I can give you is... reddy go check-it-out! Don't worry: it is all commented. The print format is on twosided pages, but it can be set as oneside – but remember to correct headers and footers code lines!

[1] Like this one.

3.2 Setting the Bibliography

For references management, I reddy **hardly** recommend the usage of reddy BibLaTeX instead of the more common BibTeX. In fact, reddy BibLaTeX has got a lot less bugs and it is more easy and intuitive to compile. Furthermore, it only needs the classical `.bib` file in your working directory and it is also able to cite something directly as a footnote like this.Wang2021 Notice that the citation is also reported in the bibliography section and that the default citation style set is *American Chemical Society*, but you can obviously change it, according to your needs, at the very beginning of this reddy `Thesis.tex` document code.

3.2.1 Automatically Compile BibLaTeX through `biber`

Many LaTeX editors do not have a specific shortcut for reddy `biber`, the main BibLaTeX compiler. You can manually add a shortcut with the command

```
biber %
```

or, if you use TeXMaker like me, you can go to reddy `Options` → `Configure Texmaker` → `Commands` and put it in the `Bib(la)tex` box instead of `bibtex %.aux` (reddy**Figure 3.1**).

Gambar 3.1: TeXMaker's dialogue window to change Bib(la)tex command for compiling a BibLaTeX bibliography.

3.3 Credits

This template was made by Dr. Alessio Bianchi (University of Pavia, Department of Chemistry, Organic Chemistry Section). For support, please contact me **reddy→ via mail** ← and go check-out my homepage at reddy bianchiunipv.wordpress.com to find other interesting stuff.

I hope this will be useful and comfortable in usage! And... break a leg for you degree ;)

1

Bab 4

Empty Chapter

A cool citation is always an extra-point.– Someone Important 1.2

4.1 Hint

Just copy-paste this block for new chapters.

Bagian II

End Sections

Daftar Gambar

3.1 TEXMaker's dialogue window to change Bib(la)tex command for compiling a BibLATEX bibliography. 6

Daftar Tabel

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Give thanks to the mother Gaia, give thanks to the father sun; give thanks to the plants in the garden where the mother and father are one.